HARDWARE × MODELS × REAL-WORLD SPEED

Intelligence. On your desk.

A field guide to capable local coding and tool-using agents.
14 hardware configurations. Seven local models + four Claude Opus reference points.

Start with the model.
Then choose memory, speed and budget.
Practical local-agent setups, from a laptop to a deskside supercomputer.

THE PORTABLE PICK
M5 Max · 128 GB
Flash-Next + MTPLX: **64 tok/s reported** at 65K context. Its n-gram table streams from SSD.

THE CAPABLE DESK-SIDE PICK
M5 Ultra · 256 GB
GLM-5.3-Flash at 44–62 tok/s with DFlash2 in oMLX. The 512 GB model adds GLM-5.3.

THE COMPACT CUDA PICK
DGX Spark · 128 GB
Flash-Next at **37–46 tok/s reported**. Two Sparks run higher-scoring GLM-5.3-Flash at **28–63** with DFlash2.

01 / The hardware-to-model map

M reported measurement · E estimate

35 tok/s gate: choose the highest-scoring fitting model whose decode range reaches ≥35 tok/s, then sort by score and speed ↑. One active request; a qualifying upper bound is not a guaranteed minimum.

DEVICE / CONFIGURATION	APPROX. USD	MEMORY INSTALLED	BANDWIDTH GB/S	AI COMPUTE TFLOPS / TOPS	BEST-FIT SHORTLIST MODEL / PRECISION	DECODE TOKENS / SEC	DECODE BOTTLENECK
MacBook Pro · M5 [H1] 10-core GPU	**$2.5k** System	**32 GB** Unified	153	**~16 TFLOPS** E FP16 · INT8 ~30 TOPS · GPU accel.	Qwen3.6-35B-A3B · 4-bit MLX oMLX 0.3.6 · ~20 GB peak	**51–60** M 1K–4K · two runs · B12	**Bandwidth** BW 57–67% · compute 2%
MacBook Pro · M5 Pro [H1] 20-core GPU	**$3.7k** System	**64 GB** Unified	307	**~36 TFLOPS** E FP16 · INT8 ~67 TOPS · GPU accel.	Qwen3.8-27B · dynamic Q4 MTPLX · native MTP drafting	**21–44** E Upper end: repetitive editing	**Bandwidth** BW 44–93% · compute 4–8%
GeForce RTX 4090 [H3]	**$2–2.5k** Card est.	**24 GB**	1,008	**661 TFLOPS** FP8	Qwen3.8-27B · Q4_K_S llama.cpp · no MTP	**38–46** M 64K → 4K · B10	**Bandwidth** BW 67–82% · compute 0.7%
Radeon AI PRO R9700 [H4]	**$1.7k** Card	**32 GB**	640	**766 TOPS** INT4 · 383 TFLOPS FP8	Qwen3.8-27B · Q4-class + MTP llama.cpp Vulkan · MTP=2	**51.8** M AMD vendor · preliminary · B8	**Bandwidth** BW 57% · compute 1.7%
GeForce RTX 5090 [H5]	**$6.4–7.4k** Card est.	**32 GB**	1,792	**1,676 TFLOPS** FP4	Qwen3.8-27B · Q4_K_M LM Studio / CUDA	**59** M 6.1K · no MTP · B3	**Bandwidth** BW 58% · compute 0.7%
Ryzen AI Max+ 395 / Halo [H2] Radeon 8060S	**$2–4k** System est.	**128 GB** GPU allocation varies	256 LPDDR5X-8000	**50 TOPS** NPU · INT8	Qwen3.8-Flash-Next · UD-IQ4_XS llama.cpp ROCm · MTP + n-gram drafting	**32–47** M 8K new code → rewrite · B7	**Overhead** BW 23–34% · compute 1%
1 × DGX Spark [H7] Single NVIDIA GB10 system	**$4.7k** System	**128 GB**	273	**500 TFLOPS** FP4	Qwen3.8-Flash-Next · NVFP4 vLLM + MTP3 · table on SSD	**37–46** M Prose → code · B5	**Mixed** BW 35–44% · compute 0.7%
MacBook Pro · M5 Max [H1] 40-core GPU	**$6k** System est.	**128 GB** Unified	614	**~70 TFLOPS** E FP16 · INT8 ~130 TOPS · GPU accel.	Qwen3.8-Flash-Next · dynamic Q4 MTPLX 2.12 + MTP · n-gram table on SSD	**63.5** M 65.5K prompt · B1	**Mixed** BW ~41% · compute 1.9%
RTX PRO 6000 Blackwell [H6] Workstation Edition	**$16–20k** Card	**96 GB**	1,792	**2,015 TFLOPS** FP4	Qwen3.8-Flash-Next · NVFP4 vLLM · n-gram table in ~100 GB host RAM	**81–88** M No MTP; 120–173 MTP · B9	**Overhead** BW 15–17% · compute 0.2%
2 × DGX Spark [H7]	**$9.4k** Systems	**256 GB** 128 GB per node	2 × 273 Distributed memory	**1,000 TFLOPS** FP4 · summed	GLM-5.3-Flash · NVFP4 / EXL3 4-bit TP2 · DFlash2 drafter · 200G RoCE	**28–63** M Prose → code · B16	**Bandwidth** BW 62–64% · compute 1.5%
Mac Studio · M5 Ultra [H8] 80-core GPU	**$12k** System est.	**256 GB**	1,200	**~140 TFLOPS** E FP16 · INT8 ~260 TOPS · GPU accel.	GLM-5.3-Flash · oQ4e / mixed 4/8-bit oMLX + DFlash2 drafter	**44–62** M 4K → short prompt · B17	**Overhead** BW 25–36% · compute 2–3%
2 × RTX PRO 6000 Blackwell [H6]	**$32–40k** Cards est.	**192 GB** 96 GB per GPU	2 × 1,792 Not one shared bus	**4,030 TFLOPS** FP4 · summed	GLM-5.3-Flash · IQ4_XS Layer split · ~157 GB weights · fit projected	**41–70** E Best ~55 · B13 anchor	**Overhead** BW 25–42% · compute 0.2%
M5 Ultra · 512 GB [H8] Ships late October	**$15–18k** System est.	**512 GB**	1,200	**~140 TFLOPS** E FP16 · INT8 ~260 TOPS · GPU accel.	GLM-5.3 · IQ4_XS / oQ4 Needs DFlash2 drafter · ~365–420 GB weights	**20–40** E Drafting · 15–25 without	**Bandwidth** BW 30–50% assumed
DGX Station · GB300 [H9] OEM order · deskside	**$99–124k** System	**748 GB** 252 HBM + 496 CPU	7,100 / 396 HBM / LPDDR5X	**15,000 TFLOPS** FP4 vendor	GLM-5.3 · IQ4_XS, split HBM / CPU Needs DFlash2 drafter + split-memory backend	**21–40** E Drafting · 16–25 without	**CPU memory** 396 GB/s tier · compute <1%

Prices: approximate USD before tax; cards exclude the host PC; "est." means a budget estimate. **Speeds:** externally reported (M) or estimated (E); model, context and runtime differ. **Compute:** dense matrix peaks of the GPU, or the NPU where labeled. Apple has no published figure, so its E values are third-party estimates for the matrix units built into each M5 GPU core. Peak compute mostly drives prefill; single-user decode is set by bandwidth, which is why the Macs decode well despite lower peaks. **Bottleneck:** share of peak memory bandwidth and FP16 compute implied by the listed speed. At ≥55% of bandwidth a pair is bandwidth-bound; below ~40%, fixed per-token overhead (launches, synchronization, host lookups) dominates.

02 / Local models & the frontier trajectory

BASELINE Claude Opus · cloud yardstick, not runnable locally — AA Intelligence Index v4.3.2 · higher is better

10 independent evaluations: agents 30% · coding 20% · scientific reasoning 20% · general 30%. **Opus trajectory:** 4.6 (Feb) 32* → 4.8 (May) 42 → 5 (Jul) 51 → 5.5 (Sep) 58. The best local-fitting model, GLM-5.3 (45), sits between May's Opus 4.8 and July's Opus 5.

MODEL / TESTED SETTING	AA SCORE	PARAMETERS / TYPE	MEMORY ESTIMATE · 128K CONTEXT · 1 ACTIVE REQUEST
Claude Opus 5.5 BASELINE Max · Sep 2026 · cloud ceiling	58	**Undisclosed** Architecture undisclosed	**Cloud only · weights and serving memory undisclosed**
Claude Opus 5 BASELINE Max · Jul 2026 · cloud reference	51	**Undisclosed** Architecture undisclosed	**Cloud only · frontier trajectory reference**
GLM-5.3 Max	45	**~753B · MoE** ~40B active	**365 GB weights + ~13 GB cache ≈ 378 GB resident** IQ4_XS · compressed MLA, 16-bit cache · SSD tables: none · runtime: +8–16 GB
Claude Opus 4.8 BASELINE Max · May 2026 · cloud reference	42	**Undisclosed** Architecture undisclosed	**Cloud only · frontier trajectory reference**
GLM-5.3-Flash Reasoning	42	**320B · MoE** 18B active	**157–200 GB weights + ~2 GB cache ≈ 159–202 GB resident** Q4-class · compressed MLA, 16-bit cache · SSD tables: none · runtime: +8–16 GB Expanded-KV backends need ~95–105 GB cache instead.
Qwen3.8-Flash-Next Reasoning	40	**~180B · MoE + lookup** 125B backbone · 6B active	**~83 GB weights + ~4 GB cache ≈ 87 GB resident** MTPLX dynamic Q4 · 16-bit cache · +32 GB lookup tables on SSD · runtime: +4–8 GB
DeepSeek V4.1 Flash Max	39	**~763B · MoE + Engram** 552B backbone · 16B active decode	**~316 GB weights + ~0.2–0.6 GB cache ≈ 317 GB resident** DwarfStar Q4 · packed FP4–BF16 cache scenarios · +203 GB Engram on SSD · runtime: +8–16 GB
Qwen3.8-27B Xhigh	34	**27B · Dense** 27B active	**20.4 GB weights + ~9.3 GB cache ≈ 30 GB resident** MTPLX dynamic Q4 · 16-bit cache · SSD tables: none · runtime: +4–8 GB
DeepSeek V4 Flash / 0731 Max · text model	34	**284B · MoE** 13B active	**~155 GB weights + ~1 GB cache ≈ 156 GB resident** Q4_K_XL · compressed attention, BF16 cache · SSD tables: none · runtime: +8–16 GB
Claude Opus 4.6 BASELINE Max · Feb 2026 · AA estimate	32	**Undisclosed** Architecture undisclosed	**Cloud only · frontier trajectory reference** AA marks 32 as estimated; independent evaluation forthcoming.
Qwen3.6-35B-A3B Reasoning · fallback	18	**35B · MoE** ~3B active	**~22 GB weights + ~3 GB cache ≈ 25 GB resident** Q4 · 16-bit cache · runtime: +4–8 GB